

Note on Commutative Algebra

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Acknowledgement

This note is basically a copy of Eisenbud's Commutative Algebra with a View Toward Algebraic Geometry, with some additional explanations and examples from other sources.

This latex style is made by Evan Chen.

1 Chapter 1: Roots of Commutative Algebra

Corollary 1.1 (Summand of polynomial rings over fields is finitely generated)

Let k be a field, and let R be a k -subalgebra of $S = k[x_1, \dots, x_n]$. R is a direct summand of S in the sense that there is a surjective R -algebra homomorphism $\phi : S \rightarrow R$ such that $\phi|_R = \text{id}_R$ and ϕ preserves degrees. Then R is finitely generated as a k -algebra.

Proof. Intuition: R itself is a graded k -algebra, with grading inherited from S . As a graded algebra, R can be decomposed into its homogeneous components. R_0 is easy to get, just include 1 in the generators. Therefore, to generate R , we really need to generate the positive degree part. That motivates us to look at the ideal R_+ .

But R_+ may not be finitely generated as an ideal. However, we know that S is a Noetherian ring, so we can look at the ideal R_+S . It must be finitely generated, say by $f_1, \dots, f_m \in R_+$.

Now, we claim that $R = k[1, f_1, \dots, f_m]$. Certainly, the right hand side is contained in the left hand side. For the other direction, it suffices to show that any homogeneous element of R can be generated by $f_1, \dots, f_m$ with coefficients in k . We prove this by induction on degree d . $d = 0$ is trivial. Now suppose we have shown this for degree less than d . Take any homogeneous element $r \in R$ of degree d . Since $r \in R_+S$, we can write $r = \sum g_i f_i$ for some homogeneous element $g_i \in S$. Apply ϕ to both sides, we get

$$r = \phi(r) = \sum \phi(g_i)\phi(f_i) = \sum \phi(g_i)f_i.$$

Note that $\deg(\phi(g_i)) = \deg(g_i) < d$, so by the inductive hypothesis, each $\phi(g_i)$ can be generated by $f_1, \dots, f_m$ over k . Therefore, r can also be generated by $f_1, \dots, f_m$ over k . This finishes the proof. $\square$

Remark 1.2. This shows that the invariant ring of a linearly reductive group acting on a polynomial ring over a field is finitely generated. If the group G is finite, the surjective homomorphism ϕ can be taken as $\phi(f) = \sum_{g \in G} gf$. (Weyl's averaging trick)

Remark 1.3. Graded rings are great because it allows us to use induction on degree. So we only need to worry about homogeneous elements.